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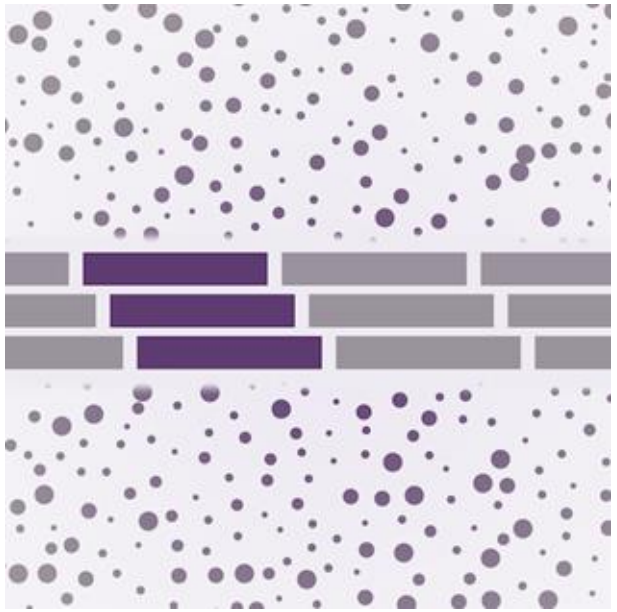
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Module 6: Structural Variants

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CBW High-Throughput Genomics

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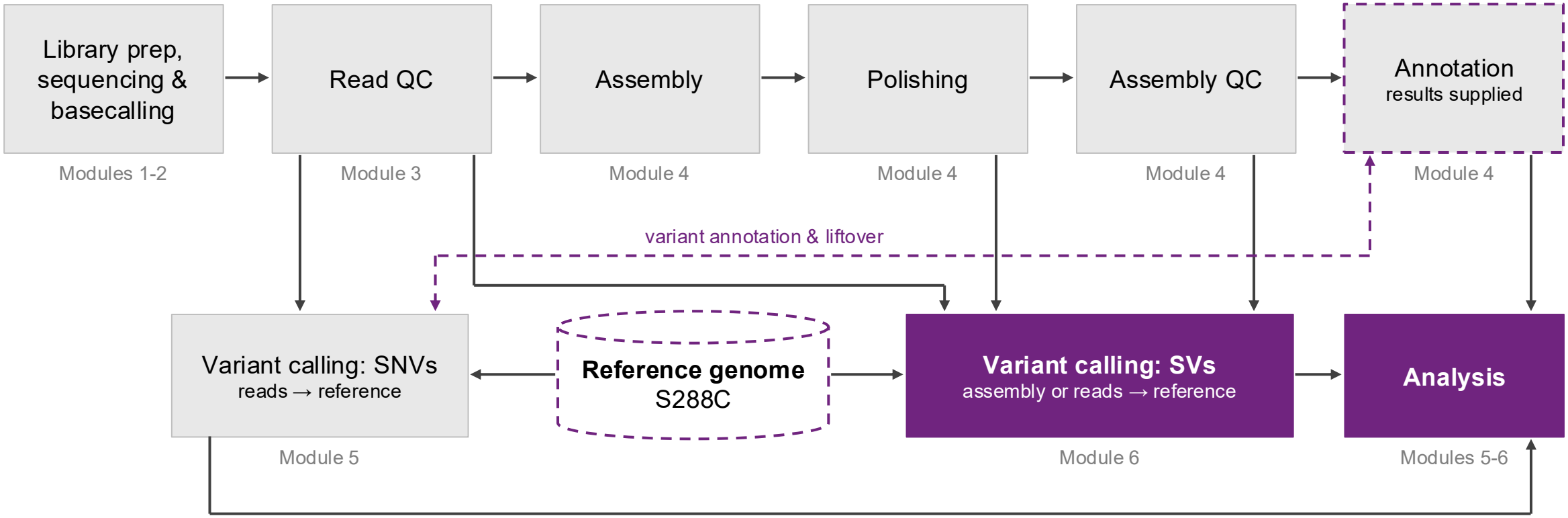


Instructor affiliation logos go here

Learning Objectives

- Define the structural variant (SV) types and explain why they matter
- Contrast SV detection from short reads, long reads and assemblies
- Interpret an SV call in a VCF and in IGV

Where we are in the workflow



= this module (dashed outline = supplied or reference)

The case: which large pieces does the wolf lack or carry?

What is a structural variant?

SNV	**Indel**	**Structural variant**
1 base substituted	1-49 bp inserted or deleted	50 bp or more



small large

Structural variant (SV): a rearrangement of **at least 50 bp**

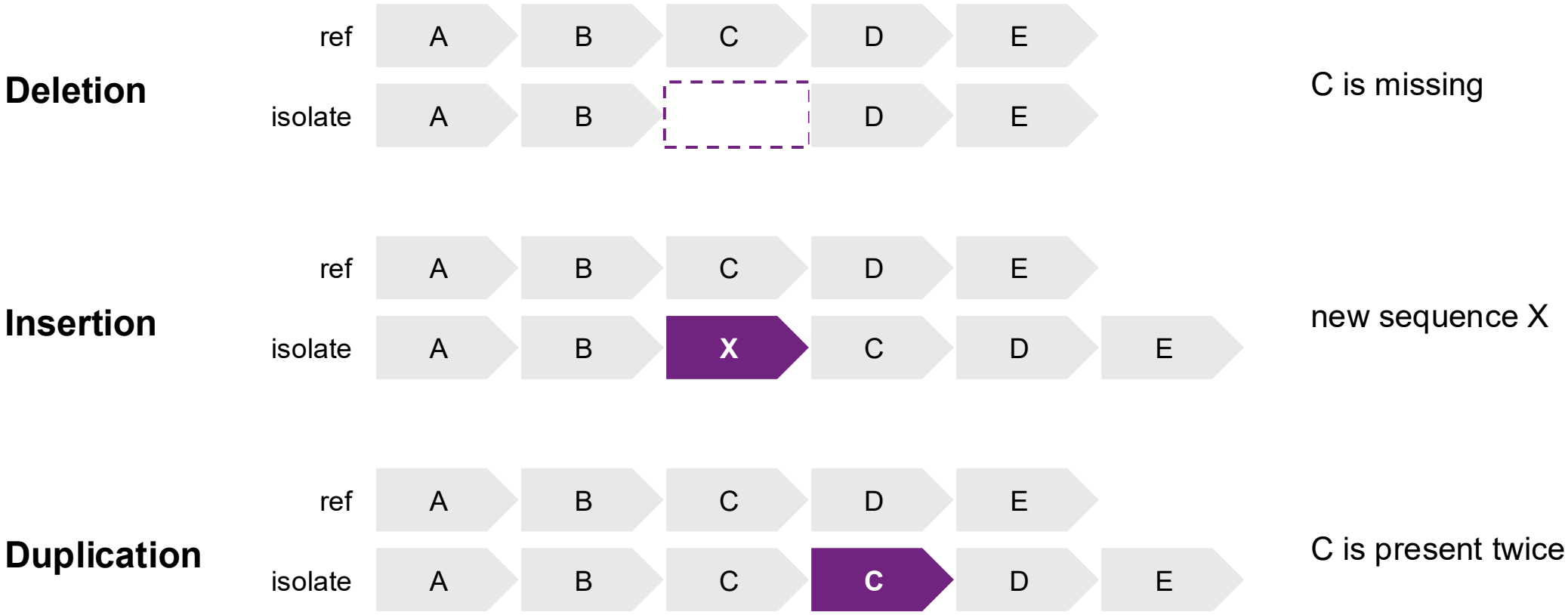
Fewer than small variants, but each changes many bases

In human genomes, SVs make up the majority of varying nucleotides (Sudmant et al. 2015)

True for the wolf (our isolate, BAL_1a)? Lab 6

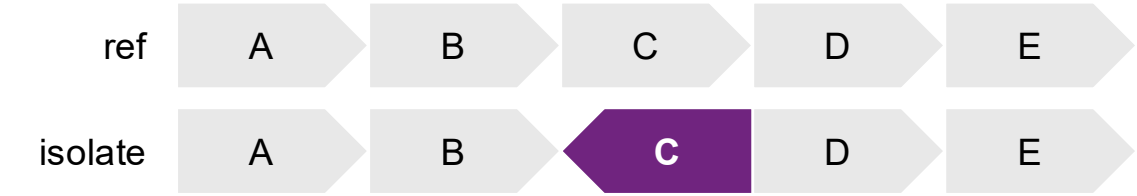
The cut-off is a convention; we use 50 bp, like the tools in the lab

Deletion, insertion, duplication



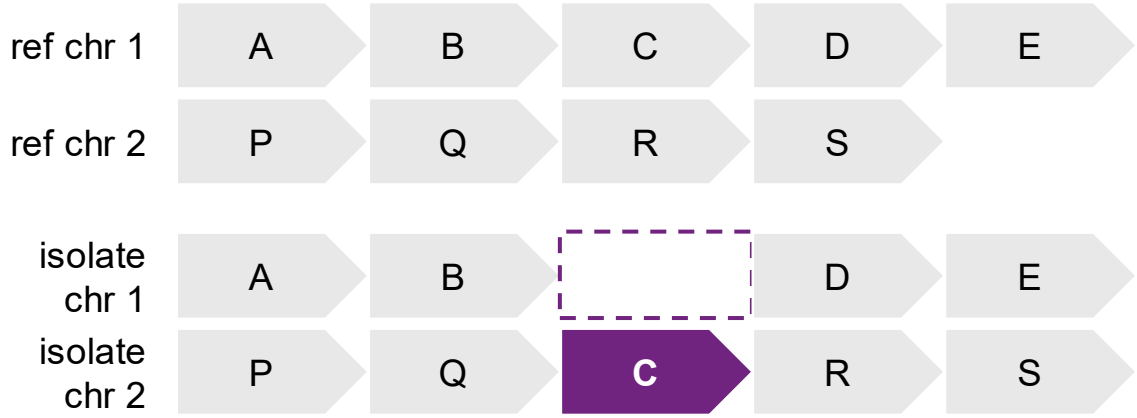
Inversion and translocation

Inversion



C is flipped

Translocation



C moved to another chromosome

Both keep the amount of DNA the same: **copy-neutral**

SV classes: copy number or not?

Group	Classes	Copy number
CNVs	Deletion, duplication	Changes
Copy-neutral	Inversion, translocation	Unchanged
Other	Novel insertion, mobile-element (Ty) insertion	Adds new sequence

CNV = copy-number variant: a deletion or duplication that changes copy number

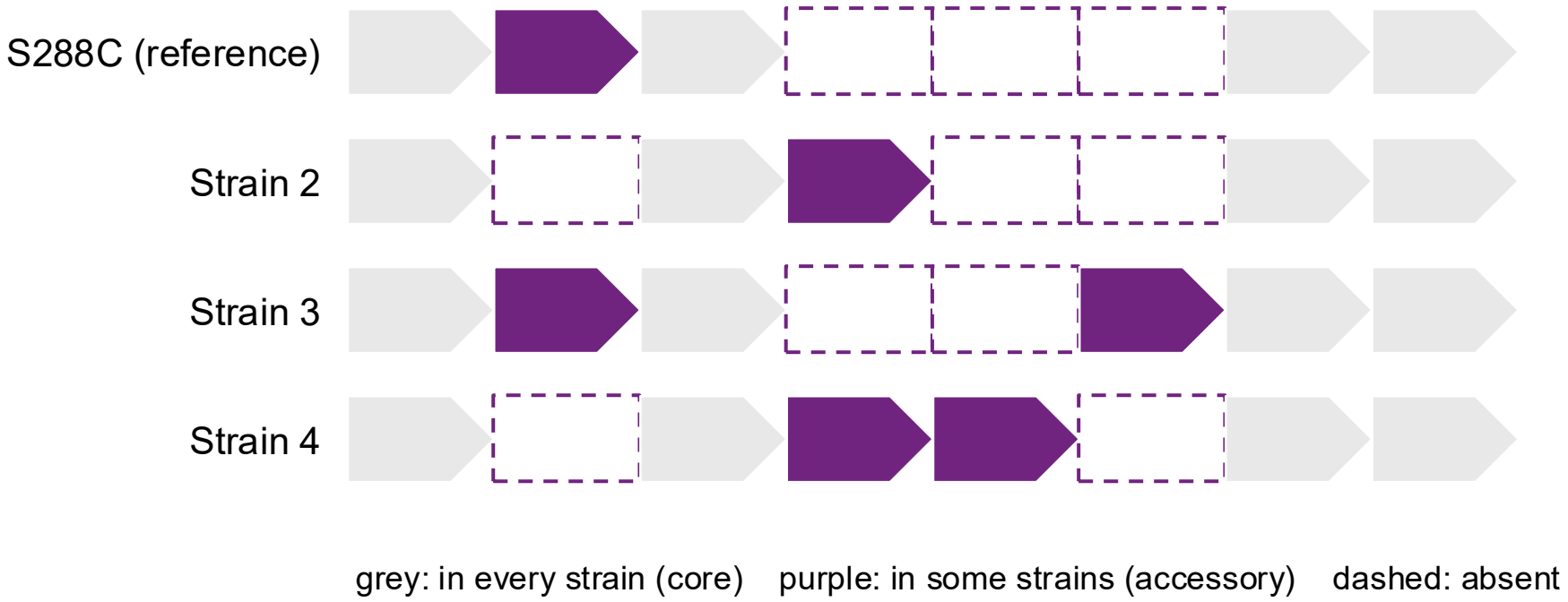
Example we meet again: the yeast **CUP1** locus, whose copy number varies between strains

Why SVs matter: genes

SV	Possible effect on genes
Deletion	Removes a gene or part of it
Duplication	More copies: more gene product (dosage)
Inversion, translocation	Breakpoint may cut a gene or move it next to new regulation
Insertion (e.g. Ty)	May land in a promoter or coding sequence

A change in the number of **CUP1** copies is associated with copper resistance

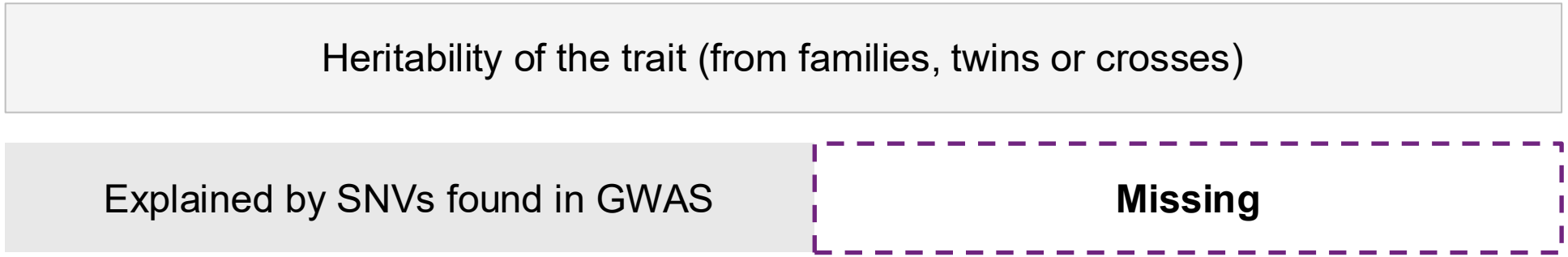
Why SVs matter: pangenomes



Pangenome: all the sequence found across many genomes of a species

Reference bias: reads from sequence the reference lacks cannot be placed

Why SVs matter: missing heritability



GWAS: scans the genome for links between variants and a trait

Part of the heritability is not explained: it is "missing"

SVs may hide in that gap: hard to see with short reads and SNV panels

They are one candidate, not the whole answer

SVs in yeast and in our isolate

ScRAP: 142 yeast strains assembled from long reads

About 4,800 non-redundant SVs reported

Our isolate, BAL_1a (the wolf), is one of the 142

Which large pieces does the wolf lack or carry? Lab 6

Four ways to find an SV

Read pair	Split read	Read depth	Assembly
Paired reads map too far apart, too close, or in the wrong orientation	One read aligns in two pieces to different places	Coverage drops or rises over a region	Build contigs, align them to the reference, read the differences

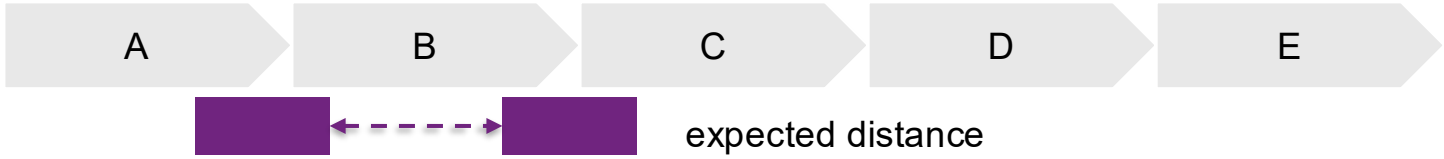
Short reads: the first three

Long reads: spanning and split alignments

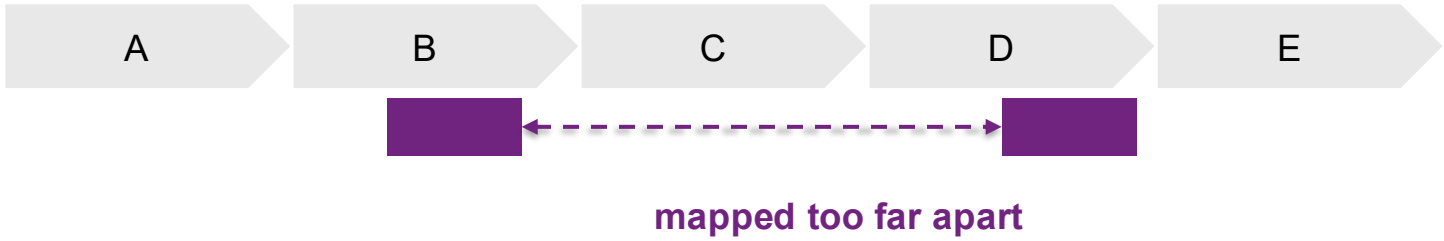
Assemblies: contigs as long reads

Read pairs: distance between mates

Concordant pair



Discordant pair: segment C is missing in the isolate



A pair at the wrong distance (too far: deletion; too close: insertion) or orientation is **discordant**

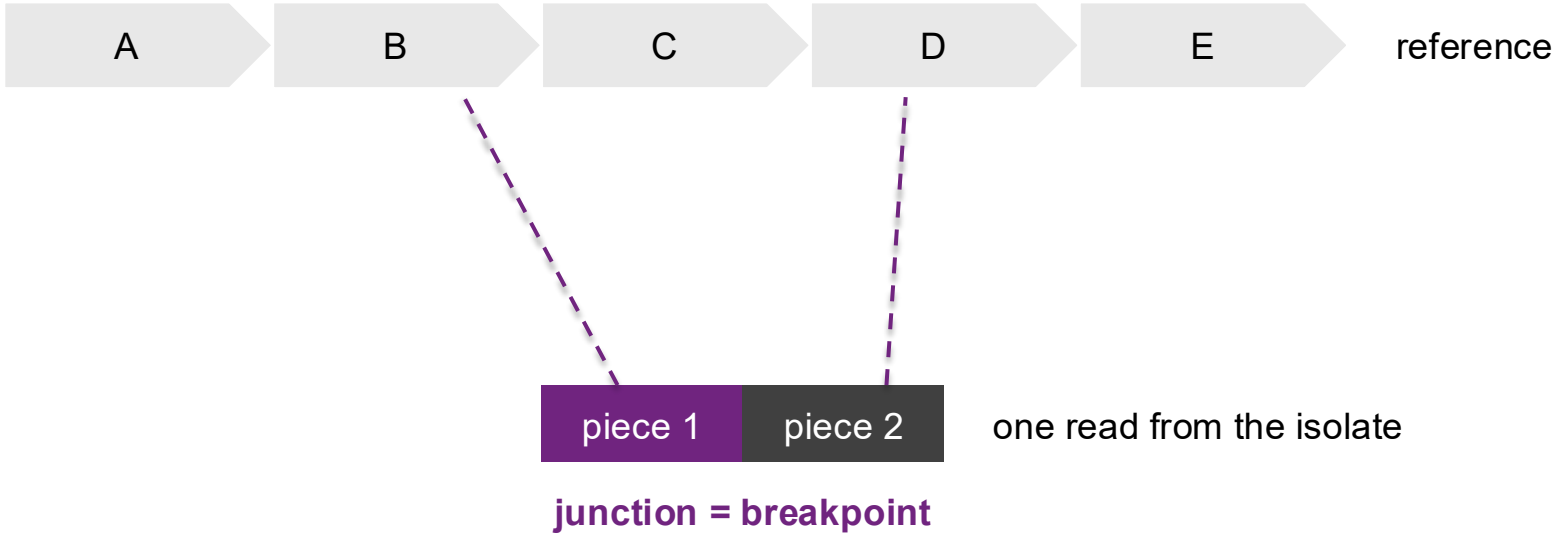
Read-pair signatures by SV class

SV class	Read-pair signal
Deletion	Mates too far apart
Insertion	Mates too close (small); one mate unplaced (large)
Inversion	Wrong orientation (same strand)
Tandem duplication	Order swapped, reads point away
Translocation	Mates on different chromosomes

Strength: most SV classes can be detected in principle

Weaknesses: repeats, false positives, breakpoint only roughly placed

Split reads cross the breakpoint



A **split read** aligns in pieces: the junction is the **breakpoint**

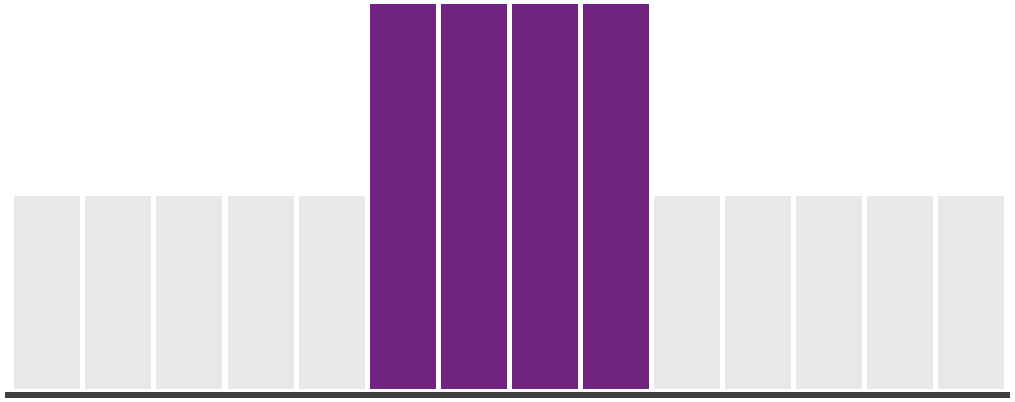
Strength: breakpoint to the base. Weakness: needs coverage; repeats

Read depth: counting reads

Deletion: depth drops



Duplication: depth rises



Depth in equal-size bins, compared with the average

Strength: gives copy number, simple. Weakness: coarse (about 5-10 kb); misses inversions and insertions

Short reads: each signal has blind spots

Signal	Strength	Weakness
Read pair	Most classes in principle	Repeats; false positives; rough breakpoints
Split read	Exact breakpoint; short events	Needs coverage; repeats
Read depth	Copy number; simple	About 5-10 kb resolution; blind to inversions

Short reads (100-300 bp) are shorter than many SVs and repeats

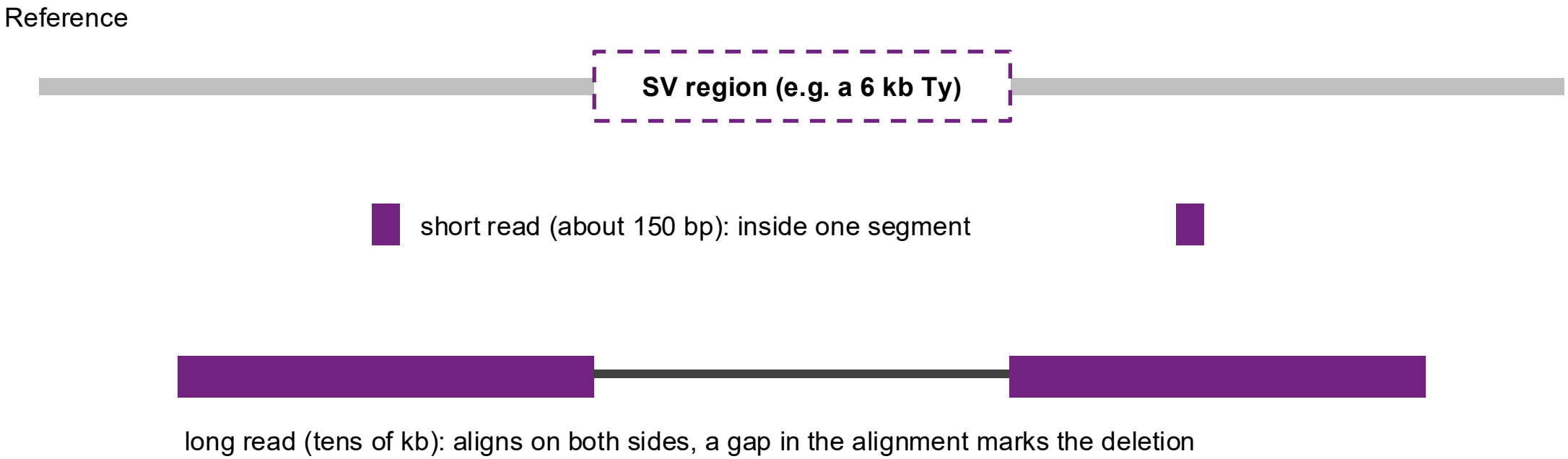
Modern callers combine signals; long reads and assemblies fix the root problem

Tools you may meet with short reads: SV calling

Tool	Signal it uses
DELLY	Read pairs plus split reads
TIDDIT	Pairs, splits and coverage
CNVpytor	Read depth; successor of CNVnator

Not covered in this workshop

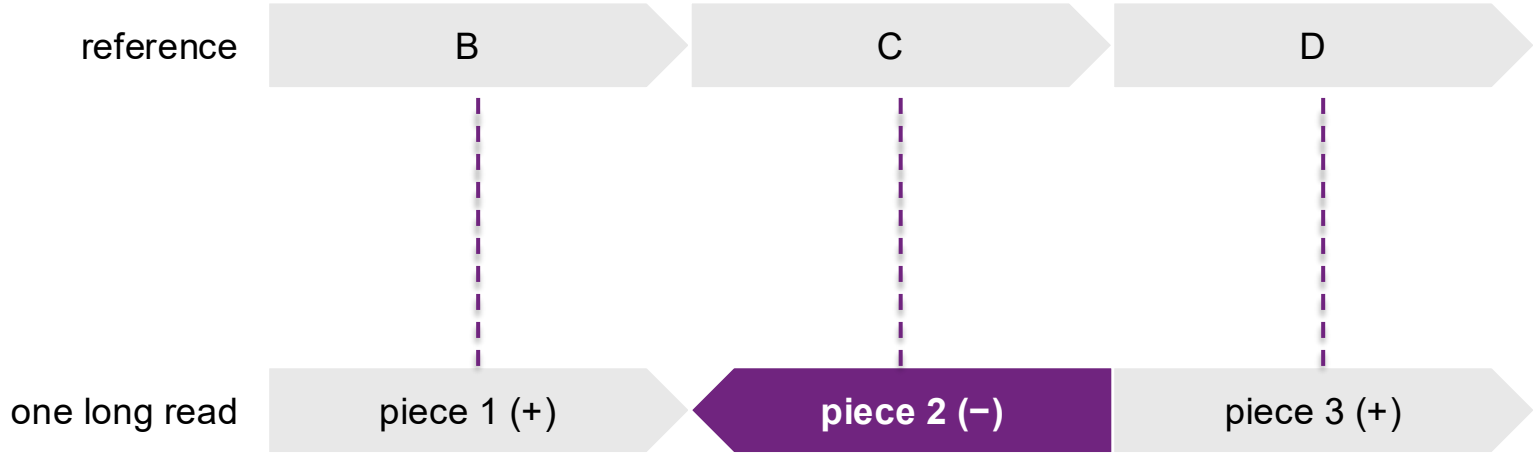
Long reads span the whole SV



A short read cannot span an SV bigger than itself: it sees a junction or an indirect signal

A **long read** can start and end in unique sequence on both sides

Long reads: split alignments



A read from an inverted segment aligns forward, reverse, forward

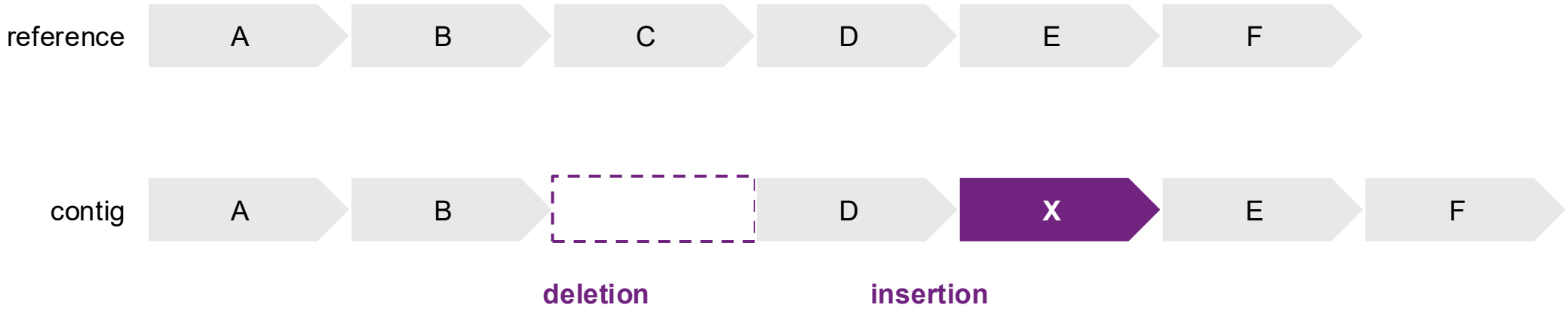
Each strand switch is a **breakpoint**; one read can cross several

Long reads: strengths and caveats

Strength	Caveat
One read can span the SV and its repeats	Reads are noisy: indel errors, homopolymers
SV seen directly in one alignment	Repeats longer than the read are still hard
Breakpoints within the read	Needs several reads that agree

In a tandem repeat, the exact **breakpoint** position can stay ambiguous

From an assembly: contigs as very long reads

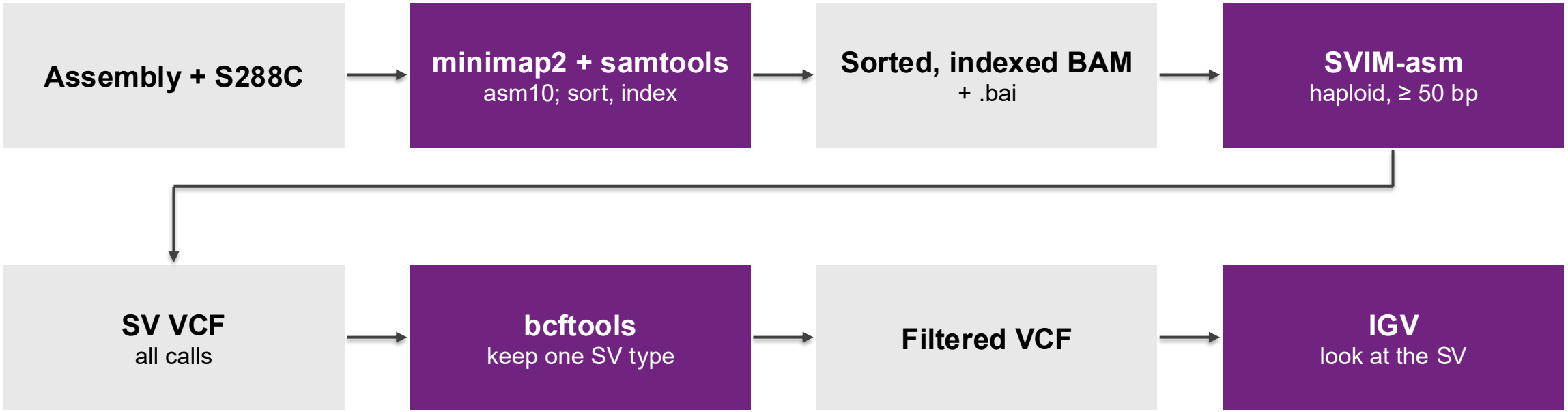


Align the contigs to the reference; read the gaps and extra sequence

Strengths: breakpoints to the base; any class, in principle

Weakness: only as good as the assembly. A **misassembly** looks like an SV

The lab route: assembly to SV calls



Alignments are SAM/BAM files, sorted and indexed with samtools (Module 5)
In the CIGAR: 3000M1500D3000M = 1,500 bp deleted; 3000M600I4000M = 600 bp inserted
A contig aligned in pieces: supplementary alignments (S = soft-clipped bases)

Purple: programs. Grey: files.

SVIM-asm and ploidy

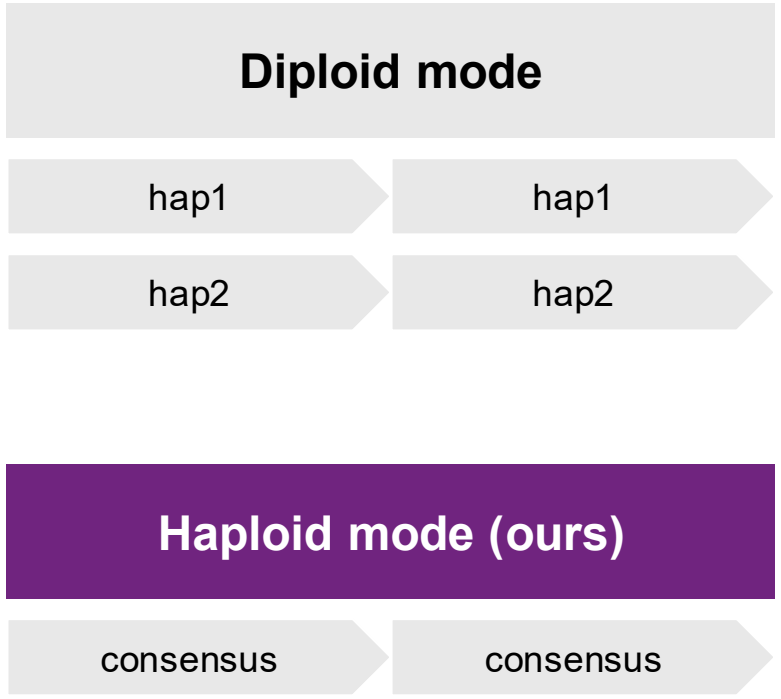
SVIM-asm reads gaps and pieces in the contig alignments

Haploid mode: one assembly. **Diploid** mode: two haplotype assemblies

Our Flye assembly is one consensus: **haploid** mode

Minimum size 50 bp, like our SV definition

Caveat: a collapsed assembly can miss an SV on one chromosome copy



Assembly-based vs read-based callers: why they can disagree

Assembly-based (SVIM-asm): reads gaps and extra sequence in contig alignments

Read-based (Sniffles2): gaps and split alignments from many reads

Not independent: the assembly was built from the same reads

In repeats, callers place breakpoints differently

Background: Ty elements

Ty1 is about 5.9 kb; a solo LTR is about 330 bp

Ty insertions cluster near tRNA genes (Kim et al. 1998)

What sizes would a Ty difference give? Lab 6

Reading an SV record: key fields

INFO field	Meaning
SVTYPE	Kind of SV: DEL, INS, DUP, INV, BND
END	Where the SV ends on the reference
SVLEN	Length in bp (negative for a deletion in these callers)
REF / ALT	Deletion: REF holds the deleted sequence. Insertion: ALT holds it
GT	SVIM-asm haploid mode: 1/1 on every call

Same VCF as module 5: only the INFO field is new

Reading an SV call (illustrative)

Caller	CHROM	POS	SVTYPE	SVLEN	END
A	IV	100,000	DEL	-600	100,600
B	IV	100,120	DEL	-595	100,715

Illustrative records, not our data. A: FILTER = PASS; ALT = C, the reference base at IV:100,000

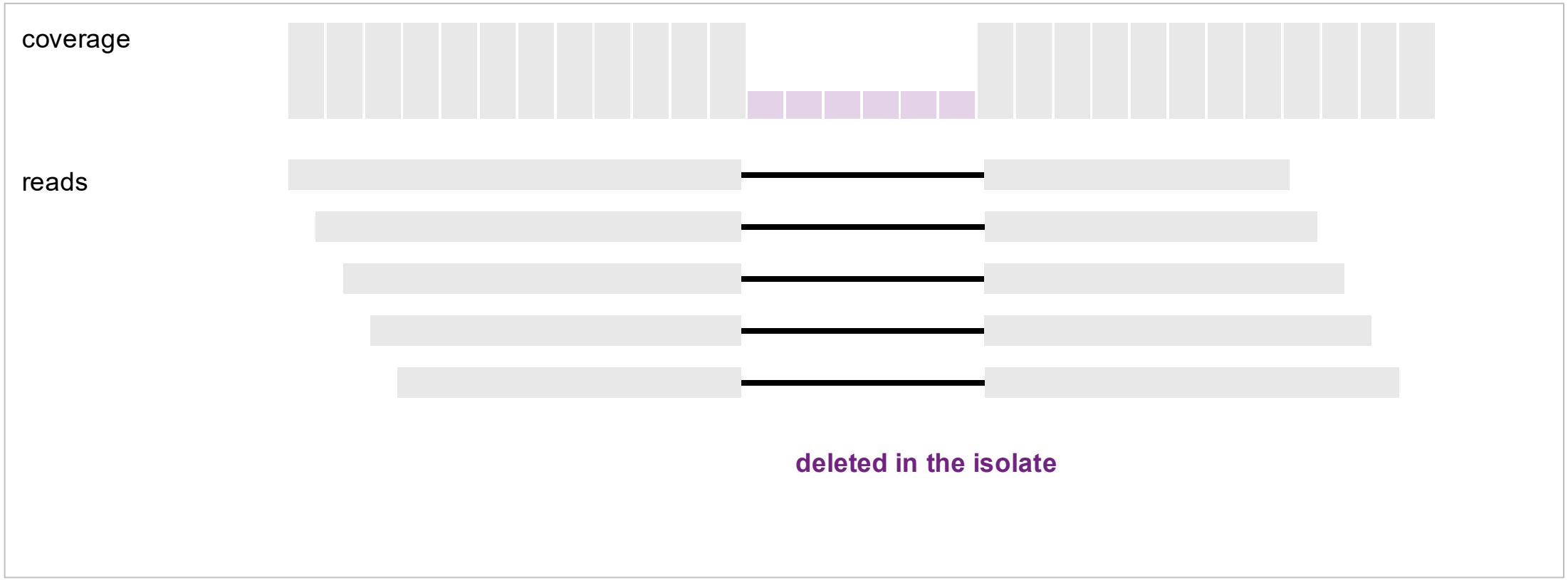
A 600 bp deletion: the reference has it, the sample does not

Same type and size; the start differs by 120 bp: one SV or two?

In a tandem repeat the breakpoint can sit anywhere within one unit

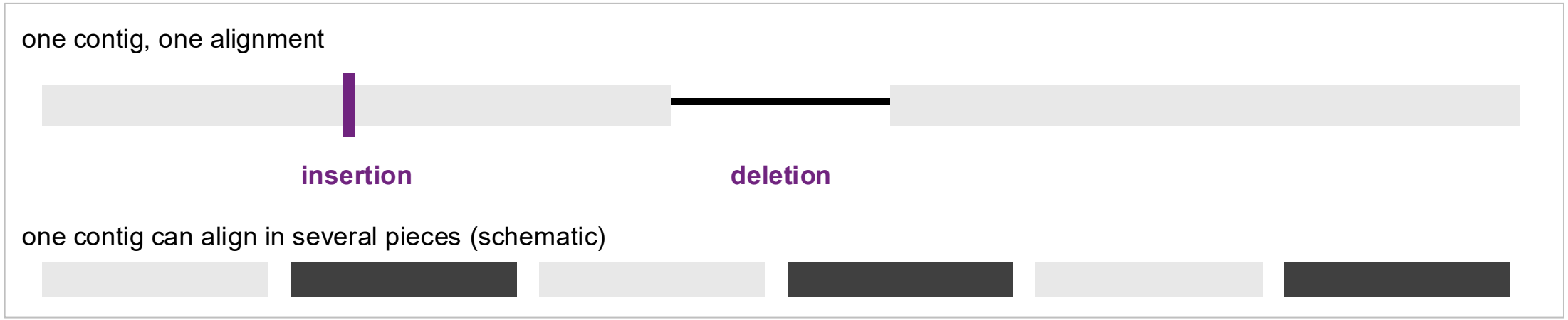
Support counts and genotypes can mislead in repeats: check the reads

A deletion in IGV: the reads



Look for: a dip in coverage, and many reads joined by a thin line across the gap
In a tandem repeat: staggered lines; coverage dips, but not to zero

SVs in an assembly BAM, and what is not an SV



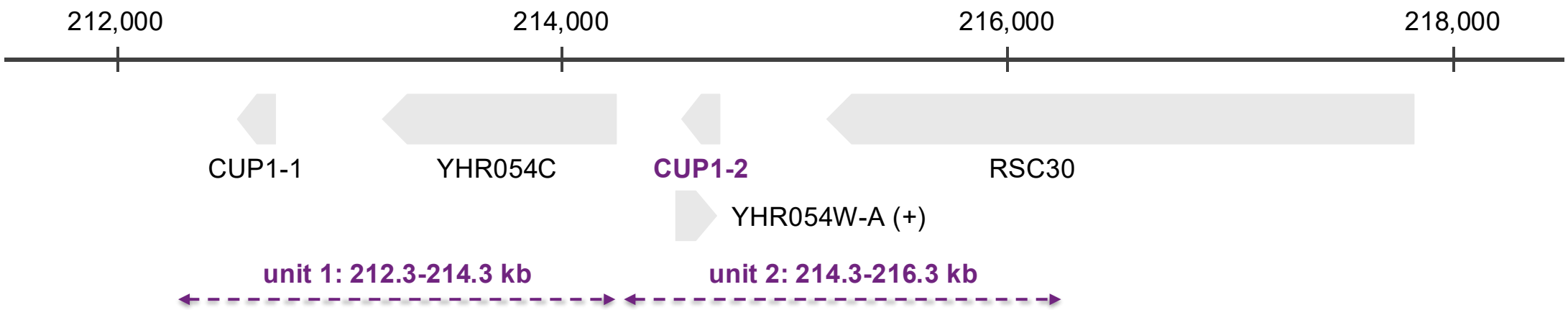
A break in the alignment is where to look, not proof of an SV

At a break the two genomes may carry different sequence there, or the assembly may be wrong

The rDNA break on chromosome XII is a contig end, not an SV (Lab 4)

Which breaks are real differences? Check the reads (Lab 6)

The stress case: CUP1 locus



S288C (the corgi): two tandem units of about 2 kb, each with a CUP1 copy

CUP1-2 (SGD S000001097): a metallothionein; copper and cadmium resistance

Variably amplified between strains; two copies in S288C

In a tandem repeat, can we tell which copy is missing? Lab 6

How many CUP1 units does the wolf carry? Lab 6 Part 4

Reading SV calls with care

- Do two methods agree? (The assembly and read routes are not fully independent)
- Is the evidence convincing in IGV: many reads, or a contig on both sides?
- Breakpoints in repeats are imprecise: callers can differ by hundreds of bp
- Near a contig end, the rDNA or a chromosome end? Be suspicious
- A misassembly can look like an SV: check the reads
- Say "in a gene associated with", not "the cause of"

Summary

- An **SV** is a rearrangement of 50 bp or more: deletion, insertion, duplication, inversion, translocation
- SVs change genes and copy number, shape the pangenome and may help explain missing heritability
- Short reads use pairs, split reads and depth; long reads span the SV; assemblies compare whole contigs
- In a VCF read SVTYPE, SVLEN and END; in IGV look for gaps and many agreeing reads
- Repeats and misassemblies can mimic or hide SVs
- Case: which large pieces does the wolf lack or carry? Lab 6

Lab 6 preview: what you will do

1	**Align the assembly to S288C** minimap2, samtools
2	**Call SVs** SVIM-asm, haploid
3	**Filter and count** bcftools: keep deletions
4	**Look in IGV** VIII:205,000-225,000
5	**Wrap up** At the end of the lab: the case review, then the survey

Coffee Break & Networking Session

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